

- C02/2/032. Let the circle $x^2 + y^2 + \alpha x + \beta y + 5 = 0$ meets the rectangular hyperbola $xy = c^2$ at four points A, B, C, D. The equation of the circumcircle of the triangle formed by joining the orthocentres of the triangles ABC, ABD and ACD is, $x^2 + y^2 + 2gx + 2f + k = 0$. If $\alpha + \beta = 6$, then the value of $(g + f + k)$ cannot be :
- (A) 2 (B) 4 (C) 5 (D) 8

Ans. BCD

Sol. The region is enclosed between two parallelograms of areas 4×2 and 2×1 .
The area $= 8 - 2 = 6$

- C02/2/033. Let $A \equiv (-2, 0)$ and $B \equiv (2, 0)$, then the number of integral values of a , $a \in [-10, 10]$ for which line segment AB subtends an acute angle at point $C(a, a + 1)$ is :
- (A) 15 (B) 17 (C) 19 (D) 21

Ans. C

Sol. Point $C(a, a + 1)$ must lie outside the circle $x^2 + y^2 = 4$.

बिन्दु $C(a, a + 1)$ वृत्त $x^2 + y^2 = 4$ के बाहर स्थित है।

$$\Rightarrow S_1 > 0 \quad \Rightarrow a^2 + (a + 1)^2 - 4 > 0$$

$$\Rightarrow a < \frac{-1 - \sqrt{7}}{2} \text{ or } a > \frac{-1 + \sqrt{7}}{2}$$

- C02/2/034. Equation of chord AB of circle $x^2 + y^2 = 2$ passing through the point $P(2, 2)$ such that $\frac{PB}{PA} = 3$ is :
- (A) $x = 3y$ (B) $x = y$
(C) $y - 2 = \sqrt{3}(x - 2)$ (D) None of these

Ans. B

Sol. $A(2 + r_1 \cos \theta, 2 + r_1 \sin \theta)$ and $B(2 + r_2 \cos \theta, 2 + r_2 \sin \theta)$
 $\Rightarrow r_1, r_2$ are roots of equation $(2 + r \cos \theta)^2 + (2 + r \sin \theta)^2 = 2$
 $\Rightarrow r_1, r_2$ मूल हैं समीकरण $(2 + r \cos \theta)^2 + (2 + r \sin \theta)^2 = 2$
 $\Rightarrow r^2 + 4(\cos \theta + \sin \theta)r + 6 = 0$

$$\Rightarrow r_1 + r_2 = -4(\cos \theta + \sin \theta) \text{ and } \text{और } r_1 r_2 = 6 \quad \text{also पुनः } \frac{r_2}{r_1} = 3$$

$$\Rightarrow \theta = \pi/4 \quad \Rightarrow y - 2 = 1(x - 2)$$

- C02/2/035. Equation of incircle of equilateral triangle ABC where $B \equiv (2, 0)$, $C \equiv (4, 0)$, is :

- (A) $x^2 + y^2 - 6x + \frac{2y}{\sqrt{3}} + 9 = 0$ (B) $x^2 + y^2 + 6x - \frac{2y}{\sqrt{3}} + 9 = 0$
(C) $x^2 + y^2 + 6x + \frac{2y}{\sqrt{3}} + 9 = 0$ (D) $x^2 + y^2 - 6x - \frac{2y}{\sqrt{3}} + 9 = 0$

Ans. AD

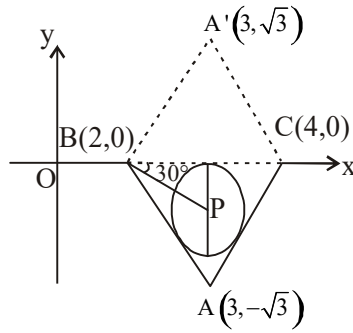
Sol.

$$\tan 30^\circ = \frac{PQ}{1}$$

$$\Rightarrow PQ = \frac{1}{\sqrt{3}} \Rightarrow r = \frac{1}{\sqrt{3}}$$

$$\Rightarrow P\left(3, -\frac{1}{\sqrt{3}}\right)$$

$$\Rightarrow (x-3)^2 + \left(y \pm \frac{1}{\sqrt{3}}\right)^2 = \frac{1}{3}$$



C02/2/036. Let C be a circle with two diameters intersecting at an angle of 30° . A circle S having radius unity, touches both the diameters and also the circle C, then the radius of circle 'C' can be :

(A) $1 + \sqrt{6} + \sqrt{2}$

(B) $1 + \sqrt{6} - \sqrt{2}$

(C) $\sqrt{6} + \sqrt{2} - 1$

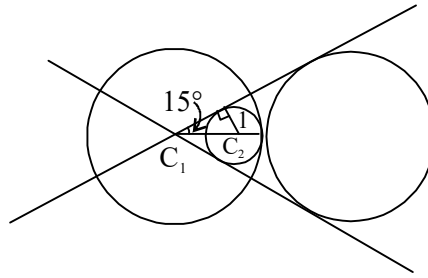
(D) $\sqrt{6} - \sqrt{2} - 1$

Ans. AC

Sol. $\operatorname{cosec} 15^\circ = \frac{C_1 C_2}{1}$

$$\Rightarrow r \pm 1 = \operatorname{cosec} 15^\circ$$

$$\Rightarrow r = \sqrt{6} + \sqrt{2} \pm 1$$



C02/2/037. Equation of the chord of the circle $x^2 + y^2 - 3x - 4y - 4 = 0$ which passes through origin such that the origin divides it in the ratio 4 : 1 is :

(A) $y = 0$

(B) $24x + 7y = 0$

(C) $7x + 24y = 0$

(D) $7x - 24y = 0$

Ans. AB

Sol. Let $y = mx$ be the chord

Points of intersection of chord and circle are given by $(1 + m^2)x^2 - (3 + 4m)x - 4 = 0$

$$\Rightarrow x_1 + x_2 = \frac{3 + 4m}{1 + m^2} \text{ and } x_1 x_2 = \frac{-4}{1 + m^2}$$

$$\text{As } x_2 = -4x_1 \Rightarrow 9 + 9m^2 = 9 + 16m^2 + 24m$$

$$\Rightarrow 7m^2 + 24m = 0 \Rightarrow m = 0, -\frac{24}{7}$$

C02/2/038. $P(4 \cos \alpha, 4 \sin \alpha)$, $Q(4 \cos \beta, 4 \sin \beta)$, $R(4 \cos \gamma, 4 \sin \gamma)$ and $S(4 \cos \delta, 4 \sin \delta)$ are four points. P_1, Q_1, R_1, S_1 respectively are their images in origin. Also $\cos \alpha + \cos \beta + \cos \gamma + \cos \delta = \sin \alpha + \sin \beta + \sin \gamma + \sin \delta = 0$,

then :

(A) orthocenter of triangle PQR lies on $x^2 + y^2 = 16$

(B) P_1 is orthocenter of triangles QRS

(C) Q_1 is circumcenter of triangle RSP

(D) R_1 is centroid of triangle SPQ

Ans. AB

Sol. Orthocentre of PQR is $(4(\cos\alpha + \cos\beta + \cos\gamma), 4(\sin\alpha + \sin\beta + \sin\gamma))$ which is $(-4\cos\delta, -4\sin\delta)$, i.e. it lies on $x^2 + y^2 = 16$

similarly P_1 is orthocentre of QRS

COMPREHENSION TYPE QUESTIONS

Paragraph for Question C01/3/039 to C01/3/041

There are two points $A(x_1, y_1)$ and $B(x_2, y_2)$ ($y_1 < y_2$) on the line $L_1 \equiv x + y = 4$. From A and B perpendicular AQ and BP are drawn to the line $L_2 \equiv 4x + 3y = 10$ such that $AQ = BP = 1$, where P & Q are the feet of the perpendiculars. On the basis of above information answer the following :

C01/3/039. The value of $\frac{y_1 + y_2}{x_1 + x_2}$ is equal to :

- (A) 3 (B) -3 (C) $-\frac{1}{3}$ (D) $\frac{1}{3}$

Ans. B

Sol. Coordinates of any point on the line $x + y = 4$ will be $\alpha, 4 - \alpha$

Now length of the perpendicular from this line to $4x + 3y = 10$ will be

$$\left| \frac{4\alpha + 3(4 - \alpha) - 10}{5} \right| = 1$$

$$\Rightarrow \alpha + 2 = \pm 5 \Rightarrow \alpha = 3, -7$$

so coordinates of A(3, 1), A(-7, 11)

so $x_1 = 3, y_1 = 1, x_2 = -7, y_2 = 11$

$$\text{so } \frac{y_1 + y_2}{x_1 + x_2} = \frac{1 + 11}{3 - 7} = -3$$

C01/3/040. The length PQ is equal to :

- (A) 8 (B) 10 (C) 12 (D) 14

Ans. C

Sol. Now foot of the perpendicular P from (3, 1) to the line $4x + 3y = 0$ will be the point of intersection of the lines $3x - 4y - 5 = 0$ and $4x + 3y = 10$

i.e. $\left(\frac{11}{5}, \frac{2}{5}\right)$

Similarly coordinates of Q will be $\left(\frac{47}{5}, -\frac{46}{5}\right)$

$$\begin{aligned}\text{So } PQ &= \sqrt{\left(\frac{47}{5} - \frac{11}{5}\right)^2 + \left(-\frac{46}{5} - \frac{2}{5}\right)^2} \\ &= \frac{1}{5} \sqrt{36^2 + 48^2} = \frac{60}{5} = 12\end{aligned}$$

C01/3/041. The obtuse angle bisector between L_1 & L_2 will be :

(A) $(5 + 4\sqrt{2})x + (5 + 3\sqrt{2})y - 10\sqrt{2}(\sqrt{2} - 1) = 0$

(B) $(5 + 4\sqrt{2})x - (5 + 3\sqrt{2})y + 10\sqrt{2}(\sqrt{2} - 1) = 0$

(C) $(5 - 4\sqrt{2})x + (5 - 3\sqrt{2})y + 10\sqrt{2}(\sqrt{2} - 1) = 0$

(D) $(5 - 4\sqrt{2})x + (5 - 3\sqrt{2})y - 10\sqrt{2}(\sqrt{2} - 1) = 0$

Ans. D

Sol. Equation of the bisectors between L_1 and L_2 will be

$$\frac{x + y - 4}{\sqrt{2}} = \pm \frac{4x + 3y - 10}{5}$$

as $a_1 a_2 + b_1 b_2 > 0$

so obtuse angle bisector can be obtained by taking (+) sign. So it equation will be

$$5x + 5y - 20 = 4\sqrt{2}x + 3\sqrt{2}y - 10\sqrt{2}$$

$$\Rightarrow (5 - 4\sqrt{2})x + (5 - 3\sqrt{2})y - 10\sqrt{2}(\sqrt{2} - 1) = 0$$

Paragraph for Question C02/3/042 to C02/3/043

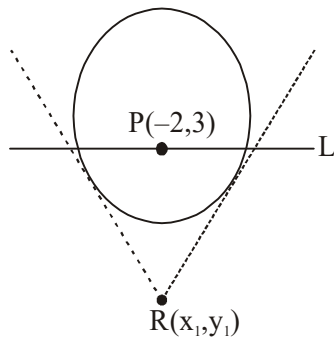
If the point P(-2,3) and Q(α , α) lie inside the circle $x^2 + y^2 = 16$ and line L is a chord of the circle for which the point P is mid point, then.

C02/3/042. The point of intersection of the tangent at the ends point of the chord 'L' :

(A) $\left(\frac{32}{13}, \frac{48}{13}\right)$ (B) $\left(-\frac{32}{13}, -\frac{48}{13}\right)$ (C) $\left(-\frac{32}{13}, \frac{48}{13}\right)$ (D) $\left(\frac{32}{13}, -\frac{48}{13}\right)$

Ans. C

Sol.



$$L : T = S_1$$

$$\Rightarrow -2x + 3y = 13 \quad \dots(1)$$

$\because$ L is c.o.c. for $R(x_1, y_1)$

$$\text{so } L : T = 0$$

$$\Rightarrow xx_1 + yy_1 = 16 \quad \dots(2)$$

$$\text{compare (1) and (2) } \frac{-2}{x_1} = \frac{3}{y_1} = \frac{13}{16}$$

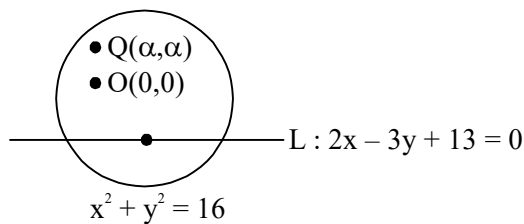
$$(x_1, y_1) \equiv \left(-\frac{32}{13}, \frac{48}{13} \right)$$

C02/3/043. Number of non-negative integral value(s) of ' α ' for which the point Q and origin lie on same side of line L is:

- (A) 3 (B) 12 (C) 13 (D) infinite

Ans. A

Sol.



$\because$ $O(0, 0)$ & $Q(\alpha, \alpha)$ lie on same side of line 'L'

$$\therefore 2\alpha - 3\alpha + 13 > 0$$

$$\Rightarrow \alpha < 13 \quad \text{_____}(1)$$

$\because$ $Q(\alpha, \alpha)$ lie inside the circle

$$\therefore \alpha^2 + \alpha^2 - 16 < 0$$

$$\Rightarrow (\alpha + 2\sqrt{2})(\alpha - 2\sqrt{2}) < 0 \quad \text{_____}(2)$$